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Paper Code : OEC-IT601A Numerical Methods

UPID : 006587

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is the rate of convergence of the Bisection method?

(II) What is the recurrence formula of Euler's method?

(III) The kind of error when 3.14 approximates π is

(IV) The degree of the interpolation polynomial of a function whose values are known at 8 points is

(V) The truncation error in composite Simpson's one third rule is of order _____

(VI) "The Gauss elimination method fails when any one of the pivotal elements is 0." - The Statement is true or false?

(VII) When Newton-Raphson's method fails?

(VIII) "Milne's Predictor-Corrector method is a single step method." - the statement is true or false?

(IX) "The error in a tabulated value does not propagate in successive differences", the statement is true or false?

(X) Geometrically the Lagrange's interpolation formula for two points of interpolation represents a_____

(XI) After being rounded off to two places of decimals the number 7.16528 becomes _____

(XII) When a matrix A can be factorized into lower and upper triangular matrices?

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Evaluate $\int_0^1 \cos x\,dx$ by Trapezoidal rule, correct up to three significant figures. [5]

3. Solve the following system of equations by Gauss-Elimination method: [5]
4x+3y=20.91
3x-y=6.94

4. Explain the geometrical interpretation of Newton-Raphson method for finding a real root of an equation. [5]

5. Define forward difference operator Δ, backward difference operator ∇ and shift operator E. Prove that $E.\Delta = \Delta.E$ [5]

6. Use Newton's divided difference formula to find *f* (8) from the following table: [5]

x:	4	5	7	10	11	13
f(x):	48	100	294	900	1210	2028

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Solve the following system of equations by LU factorization method: [8]

$$x + 3y + z = 9$$
$$x + 4y + 2z = 3$$
$$x + 2y - 3z = 6$$

(b) Solve the following system of equations by Gauss-Elimination method: [7]

$$x + y + z = 9$$
$$x - 3y + 4z = 13$$
$$3x + 4y + 5z = 40$$

8. (a) Explain the convergence of Bisection method for finding a real root of an equation. [5]

(b) Find the positive real root of $x \tan x = 1.28$ using Bisection method. [5]

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(c) Find the positive real root of $x^3 = 18$ using Bisection method of 4 iterations. [5]

9. (a) Solve the following IVP using Euler's method : $\frac{dy}{dx} = x^2 + y$ with $y(0)=1$. Compute the first five steps of the solution with $h=0.1$. [7]

(b) Solve the equation $\frac{dy}{dx} = \frac{1}{x+y}$, $y(0) = 1$ for $y(0.1)$ and $y(0.2)$ using Runge-Kutta method of order 4. [8]

10. (a) What is the difference between interpolation and extrapolation? Give suitable examples. [3]

(b) Calculate f(1.135) using suitable formula: [7]

x:	1.140	1.145	1.150	1.155	1.160	1.165
f(x):	0.131030	0.13541	0.13976	0.14410	0.14842	0.15272

(c) [5]

Given the function $y = \frac{1}{x}$, Show that the divided difference of nth order is

$$y[x_0, x_1, x_2 \ldots\ldots x_n] = (-1)^n / (x_0 x_1 x_2 \ldots\ldots x_n).$$

11. (a) Solve the following system of equations by Gauss-Elimination method: [5]

$$x + 2y + z = 0$$
$$2x + 2y + 3z = 3$$
$$-y - 3z = 2$$

(b) Find the inverse of the matrix by Matrix inversion method: $A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ [4]

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(c) Solve the following system of equations by LU factorization method: [6]

$$8x_1 - 3x_2 + 2x_3 = 20$$
$$4x_1 + 11x_2 - x_3 = 33$$
$$6x_1 + 3x_2 + 12x_3 = 36$$

*** END OF PAPER ***

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